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Paddle with extendable second blade

Abstract

A self-contained paddle can be adjusted from a single paddle form with a handle on one end to a double paddle configuration having paddle blades on both ends. The paddle of the present invention allows for the handle end of a water recreation paddle to transform to a second paddle, forming a double paddle without bring a separate attachment during an outing and without interfering with the use of the paddle in the single paddle configuration when the paddle is in the single paddle configuration.

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Background/Summary

FIELD OF THE INVENTION
(1) The present invention generally relates to paddles for water recreation. In particular, the present invention is directed to a water recreation paddle with an extendable second blade.
BACKGROUND
(2) When paddling on watercraft, such as paddleboards, a user is often standing and so requires a relatively long-handled paddle with a handle on one end and a blade on the other.
SUMMARY OF THE DISCLOSURE
(3) A paddle having a single paddle configuration and a double paddle configuration includes a shaft having a first end and a second end, a first blade on the first end of the shaft, a handle on the

second end of the shaft when the paddle is in the single paddle configuration, an inner shaft having a shaft end and a handle end, wherein the inner shaft is attached at the handle end to the handle and contained within the shaft when the paddle is in the single paddle configuration, and a second blade attached to the inner shaft and having a compact configuration and an extended configuration. The handle is disconnected from the shaft and the inner shaft is extended outwardly from and connected at the shaft end to the shaft when the paddle is in the double paddle configuration and the second blade is in the extended configuration when the paddle is in the double paddle configuration.

(4) Additionally or alternatively, the inner shaft includes an opening running along a side and a support rod is attached within the inner shaft.

(5) Additionally or alternatively, the second blade includes a first edge attached to the support rod, a second edge attached to an edge support member, and a support member pivotably attached to the edge support member on one end and releasably attached to the inner shaft to secure the second blade in the extended configuration.

(6) Additionally or alternatively, the second blade further includes woven fabric and a plurality of stays.

(7) Additionally or alternatively, the edge support member is pivotably attached to the support rod.

(8) In another aspect of the invention, a paddle having a single blade configuration and a double blade configuration includes an extendable shaft having a first segment and a second segment, wherein a portion of the second segment is configured to slidably fit within the first segment, a first blade on a first end of the first segment, a handle on a first end of the second segment when the paddle is in the single blade configuration, an inner shaft having a shaft end and a handle end, wherein the inner shaft is attached at the handle end to the handle and contained within the second segment when the paddle is in the single blade configuration, and a second blade attached to the inner shaft and having a compact configuration and an extended configuration. The handle is disconnected from the first end of the second segment and the inner shaft is extended outwardly from and connected at the shaft end to the second segment when the paddle is in the double blade configuration and the second blade is extendable to the extended configuration when the paddle is in the double blade configuration.

(9) Additionally or alternatively, the inner shaft includes an opening running along a side and a support rod is attached within the inner shaft.

(10) Additionally or alternatively, the second blade includes a first edge attached to the support rod, a second edge attached to an edge support member, and a support member pivotably attached to the edge support member on one end and releasably attached to the inner shaft to secure the second blade in the extended configuration.

(11) Additionally or alternatively, the second blade further includes woven fabric and a plurality of stays.

(12) Additionally or alternatively, the edge support member is pivotably attached to the support rod.

(13) In another aspect of the invention, a method for converting a single blade paddle to a double blade paddle includes detaching a handle from a first end of a shaft of a paddle, wherein the paddle includes a first blade on a second end, extending the detached handle outward from the shaft such that an inner shaft contained within the shaft and attached to the handle on a first end is exposed, securing the inner shaft in an extended position, unfurling a second paddle from a compact configuration within the inner shaft to an extended configuration, and securing a first end of a support member to the inner shaft. A second end of the support member is attached to a first end of a second support member, a second end of the second support member is attached to the inner shaft, a first edge of the second paddle is attached to the inner shaft and a second edge of the second paddle is attached to the second support member.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) For the purpose of illustrating the invention, the drawings show aspects of one or more embodiments of the invention. However, it should be understood that the present invention is not limited to the precise arrangements and instrumentalities shown in the drawings, wherein:

(2) FIG. **1**A is a perspective view of a paddle in a single paddle configuration in accordance with an embodiment of the present invention;

(3) FIG. **1**B shows the paddle of FIG. **1** in a double paddle configuration in accordance with an embodiment of the present invention;

(4) FIG. **2**A is a detailed view of the second blade portion of the paddle of FIG. **1**B in which the handle is extended and the blade is stowed;

(5) FIG. **2**B is the detailed view of the portion shown in FIG. **2**A in which the second blade is unfurled and locked in position;

(6) FIG. **2**C is the detailed view of the portion shown in FIG. **2**A in which the second blade is unfurled and locked in position; and

(7) FIG. **3** is an exploded, detailed view of the portion shown in FIG. **2**A in which the blade webbing is removed.

DESCRIPTION OF THE DISCLOSURE

(8) A user on a recreation watercraft such as a paddleboard may switch from a standing position to a kneeling or sitting position, depending on the conditions, for example. In a standing position, the user paddles with a long paddle having a handle on one end and a blade on the other end. In a seated or kneeling position, such a paddle can be cumbersome and inefficient to use. In that situation, a double paddle would be more desirable to use. However, for many users on most recreational watercraft that are mostly paddled, stowing two paddles is undesirable. Similarly, bringing along a double paddle attachment may be inconvenient because, if remembered, it can fall off the watercraft (or would have to be secured in some way that could interfere with the use of the watercraft).

(9) Thus, there is a need for a self-contained paddle that can be adjusted from a single paddle form with a handle on one end to a double paddle configuration having blades on both ends. The paddle of the present invention allows for the end of the shaft having the handle to transform to a second blade, forming a double paddle without the need to bring a separate attachment during an outing and without interfering with the use of the paddle in the single paddle configuration when the paddle is in the single paddle configuration.

(10) The paddle includes a single paddle configuration as shown in FIG. **1**A. Paddle **100** includes a shaft **102** having a lower shaft **103** and an upper shaft **105**. Lower shaft **103** and upper shaft **105** may be slidably engaged to allow for the adjustment of the overall length of shaft **102**. A first blade **104** is on one end of lower shaft **103** and a handle **108** on one end of upper shaft **105**. Handle **108** may be loosened from the end of upper shaft **105** and extended outwardly revealing an embedded inner shaft **112** which has a second blade **116** attached in a compressed state (as shown in FIG. **2**A) that can be placed in an unfurled state to form a double paddle configuration as shown in FIG. **1**B. Inner shaft **112** may be cylindrical with an opening running along one side, such as a ½ or ¾ cylinder (as best seen in FIG. **2**A), which houses a folded or compressed second blade **116** when in a stowed position, as when inner shaft **112** is within upper shaft **105**. Inner shaft **116** may be secured in the extended position by any suitable mechanism, such as a hole **120** for receiving a depressible securing knob **124**. Once inner shaft is extended and secured in the extended position, second blade **116** may be unfurled and secured into a blade form. Preferably, a rod **114** or other support member may be attached to an inner portion of inner shaft **112**, such as by rivets **121** (e.g., **121**A-**121**C as shown in FIG. **3**). In an embodiment, second blade **116** may be made of woven fabric or other suitable material with a plurality of supports **128** (e.g., **128**A-**128**B) at intervals for added rigidity.

(11) Turning to FIGS. **2**B-**3**, a first edge **115** of second blade **116** is attached, such as by a sewn seam, to rod **114** and a second edge **117** is attached, such as by a sewn seam, to an outer edge support member **132**. Outer edge support member **132** is pivotably attached to rod **114** or inner shaft **112** at a pivot point **44** and may include a pivotably attached blade support **136** that is attached on one end to the distal end of outer edge support member **132** and can be pivoted around pivot point **119** so that blade support **136** extends along an outside edge **113** of second blade **116** and engages, via a female snap **142** or other suitable mechanism, with a male snap **140**, socket, or other suitable receptor on inner shaft **112** that can reversibly secure the other end of blade support **136** in position to maintain second blade **116** in an unfurled configuration. Other securement mechanisms may be used to maintain second blade **116** in a blade shape for use when paddle **100** is in the double paddle configuration.
(12) In this way, a single ended long paddle for a "standup" paddleboard has an extendable shaft with an enclosed second blade that can be deployed so that the paddle becomes a double paddle. Note that the shaft of the paddle may also be adjustable in length, which is beneficial for both single paddle uses and double paddle uses.
(13) The adjustable handle at the end of the single paddle configuration allows for standard use of single long paddle with handle, which can be converted to a double ended paddle by unlocking shaft and extending secondary blade, which was folded inside shaft. Once secured in place this would allow for double ended use. To transform back to a single paddle configuration, the securement mechanism would be released and the flexible material folded back into main shaft and the handle resecured in place at the end of the shaft.
(14) The folded webbing used for the collapsible and stowable secondary blade may be made of various materials in addition to a woven fabric such as rip stop nylon or a plastic with a frame and support stays of aluminum or any other suitable light and supportive material.
(15) Exemplary embodiments have been disclosed above and illustrated in the accompanying drawings. It will be understood by those skilled in the art that various changes, omissions, and additions may be made to that which is specifically disclosed herein without departing from the spirit and scope of the present invention.

Claims

1. A paddle having a single paddle configuration and a double paddle configuration comprising: a shaft having a first end and a second end; a first blade on the first end of the shaft; a handle on the second end of the shaft when the paddle is in the single paddle configuration; an inner shaft having a shaft end, a handle end, and an opening running along a length between the shaft end and the handle end, wherein the inner shaft is attached at the handle end to the handle and contained within the shaft when the paddle is in the single paddle configuration; and a second blade attached to the inner shaft and having a compact configuration and an extended configuration, wherein the second blade fits within the inner shaft when in the compact configuration, wherein, when the paddle is in the double paddle configuration, the inner shaft extends outwardly from and is connected at the shaft end to the shaft, and the second blade extends outwardly from the opening of the inner shaft to form the extended configuration.
2. The paddle of claim 1, wherein a support rod is attached within the inner shaft.
3. The paddle of claim 2, wherein the second blade includes a first edge attached to the support rod, a second edge attached to an edge support member, and a support member pivotably attached to the edge support member on one end and releasably attached to the inner shaft to secure the second blade in the extended configuration.
4. The paddle of claim 3, wherein the second blade further includes woven fabric and a plurality of stays.
5. The paddle of claim 3, wherein the second blade further includes nylon material.

6. The paddle of claim 3, wherein the edge support member is pivotably attached to the support rod.
7. A paddle having a single blade configuration and a double blade configuration comprising: an extendable shaft having a first segment and a second segment, wherein a portion of the second segment is configured to slidably fit within the first segment; a first blade on a first end of the first segment; a handle on a first end of the second segment when the paddle is in the single blade configuration; an inner shaft having a shaft end and a handle end, wherein the inner shaft is attached at the handle end to the handle and contained within the second segment when the paddle is in the single blade configuration; and a second blade attached to the inner shaft and having a compact configuration and an extended configuration, wherein the second blade fits within the inner shaft when in the compact configuration, wherein, when the paddle is in the double blade configuration, the inner shaft extends outwardly from and is connected at the shaft end to the second segment, and the second blade extends outwardly from the inner shaft to form the extended configuration while the handle remains attached to the handle end of the inner shaft.
8. The paddle of claim 7, wherein the inner shaft includes an opening running along a side and a support rod is attached within the inner shaft.
9. The paddle of claim 8, wherein the second blade includes a first edge attached to the support rod, a second edge attached to an edge support member, and a support member pivotably attached to the edge support member on one end and releasably attached to the inner shaft to secure the second blade in the extended configuration.
10. The paddle of claim 9, wherein the second blade further includes a woven fabric material and a plurality of stays.
11. The paddle of claim 9, wherein the edge support member is pivotably attached to the support rod.
12. A method for converting a single blade paddle to a double blade paddle comprising: detaching a handle from a first end of a shaft, wherein the shaft includes a first blade on a second end of the shaft; extending the detached handle outward from first end of the shaft such that an inner shaft contained within the shaft and attached to the handle on a first end of the inner shaft is in an extended position; securing, at the first end of the shaft, the inner shaft in the extended position; and unfurling a second blade from a compact configuration within the inner shaft to an extended configuration.
13. The method of claim 12, wherein unfurling the second blade includes pivoting a first end of a first support member outwardly from the inner shaft, wherein the first support member is attached to a first edge of the second blade, and securing a first end of a second support member to the inner shaft, wherein the second support member is attached to a second edge of the second blade and pivotably attached to a second end of the first support member.